

CSC-335

HCI & Computer

Graphics

Lec 8: UX Design Concept to Wireframe



MileStone 2



Steps & Activities for M2

Step	Activities	Output
Conduct needs finding and analysis	• Interviews • Observations • Affinity diagramming • Synthesis and prioritization	Needs finding report
Sketch and ideate	• Sketching • More sketching • Brainstorming • Selecting and refining	Design concept Storyboards
Create personas and scenarios	• Identify key user groups • Identify key tasks • Creative writing	Personas and Scenarios
Do a competitive analysis	• Search web/App Stores for similar products • Identify key competitors • Map competitor functionality • Identify competitor target users • Investigate user reception (product reviews, forums)	Competitive matrix Design recommendations

UX Design: From Concept to Wireframe



Design Problem

- **Imagine you need to design a new mobile app to help with weight loss**
- **How would you figure out what this app should do and how it should do it?**



Wicked Problems...

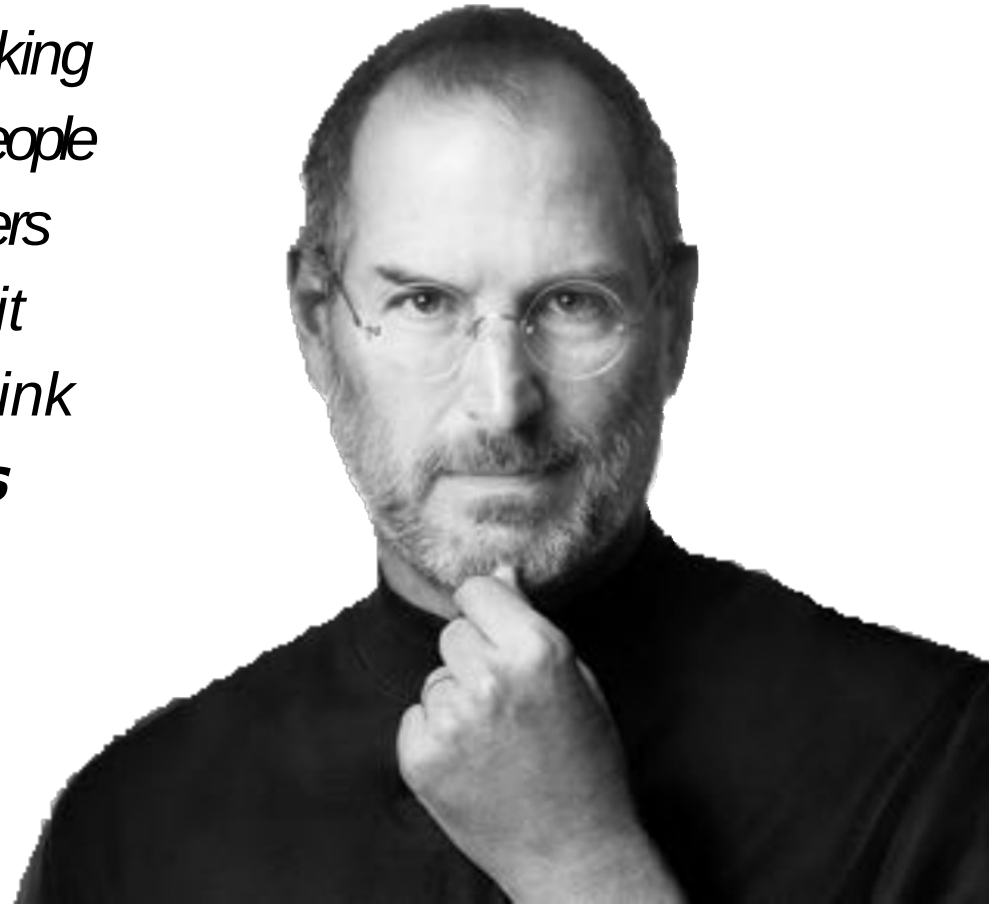
- **Are ill-defined / underspecified**
- **Don't have a right or wrong solution**
- **Are context-dependent (and, so, unique)**
- **Don't have a clear test for solutions**

Recap: What is Design?

- *Most people make the mistake of thinking design is **what it looks like**. People think it's this veneer – that the designers are handed this box and told, “Make it **look good!**” That's not what we think design is. It's **not** just what it **looks like and feels like**.*

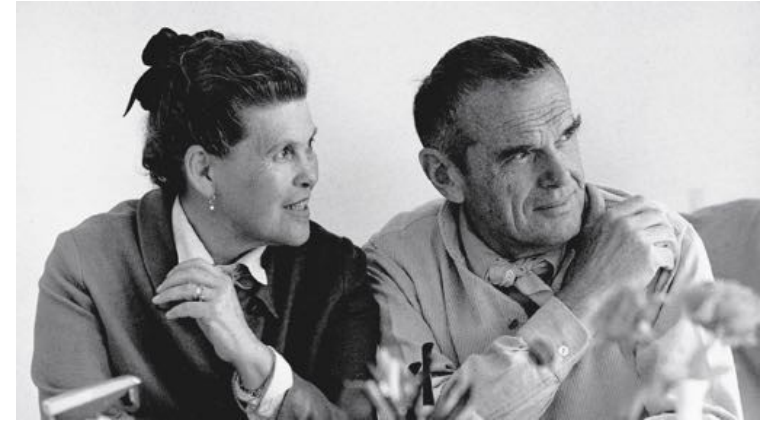
*Design is how it **works**.*

- Steve Jobs



Recap: What is Design?

- **[Design is] a plan for arranging elements in such a way as to best accomplish a particular purpose**
 - *Charles Eames*





Purpose

What is the function of this object?

What is Design?

- **...every design problem begins with an effort to achieve fitness between two entities: the form in question and its context.**

- *Christopher Alexander*





Context

Context: in what situations does this thing need to be used?

Design

- **Design is
Compromise**
- **- Bill Buxton**





Constraints

*what determines how this thing
can be made and used?*



Design

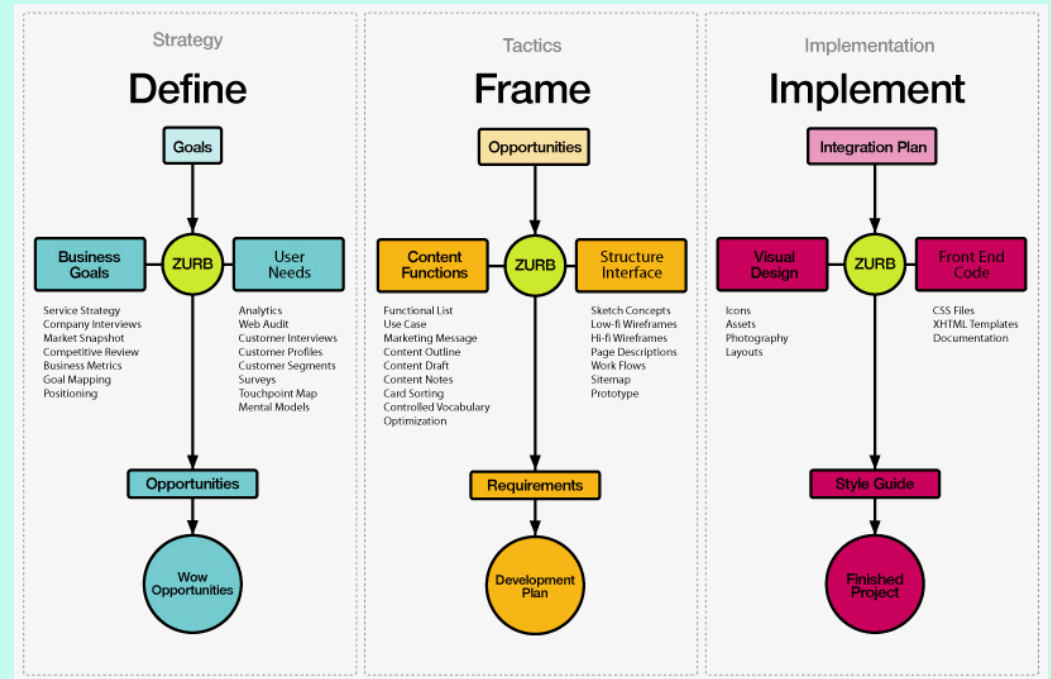
*Design is about
creating things that
fulfill their purpose
well, given the
constraints.*

Core Design Skills

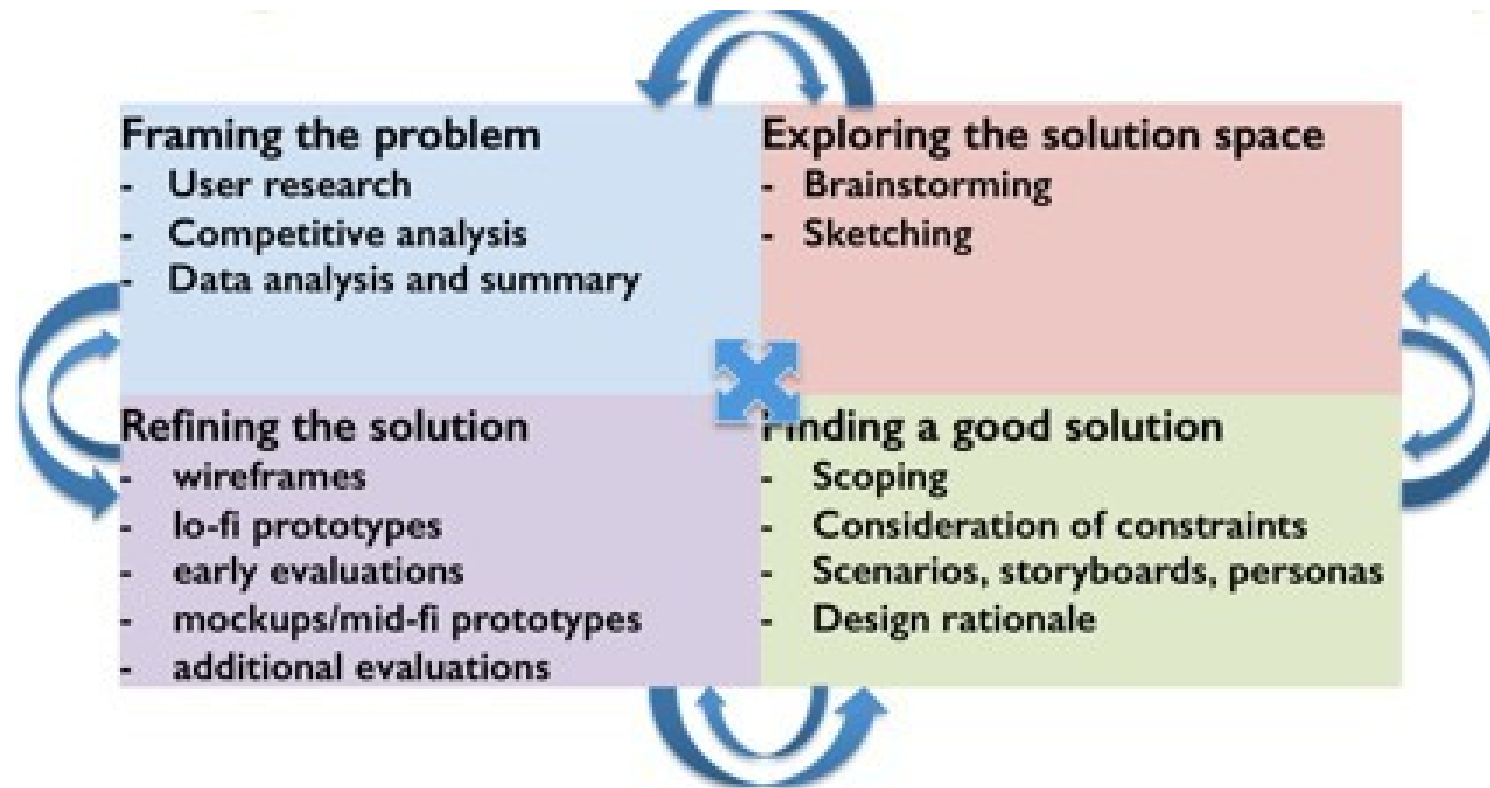
- **To frame, or reframe, the problem and the objective**
- **To create and envision alternatives**
- **To select from those alternatives**
- **To visualize and prototype the intended solution**
- **To synthesize a solution that addresses all important constraints**

Design

Is a Process



Design Process (In a Nutshell)





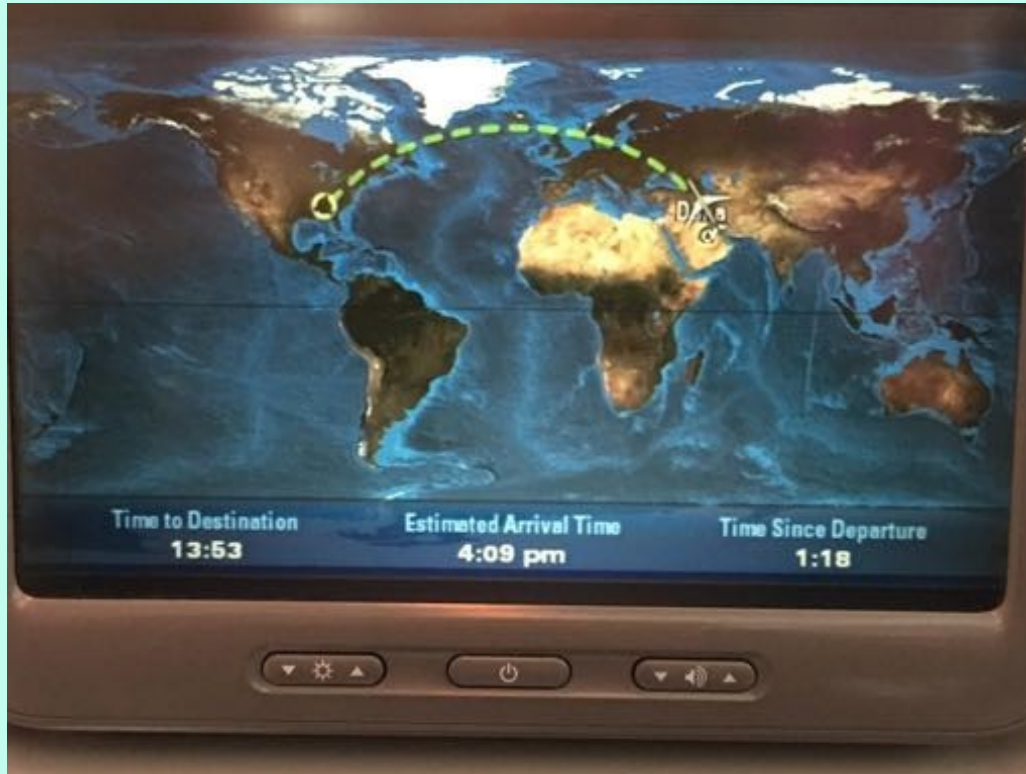
Conducting Formative Research to Frame your Design

Understand the existing context and domain

+ identify targets for design improvements

Framing the Design Problem



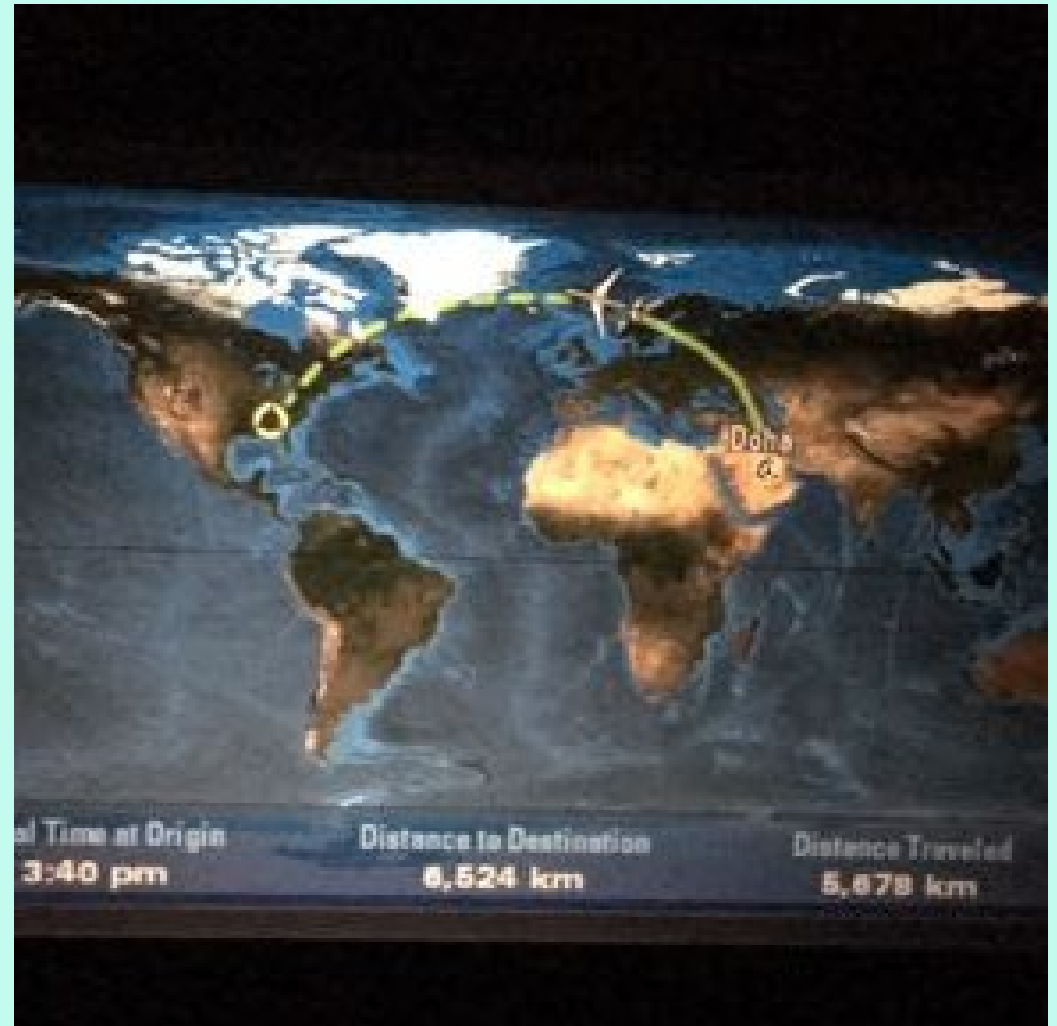


Goal

Formulate the problem and understand important constraints a solution needs to satisfy

Frame

A frame is an active perspective that both describes and perceptually changes a given situation





Whats In and Whats Out

Frame describes aspects of the design problem the designer considers to be important and will focus on.

Scope

How “big” is the problem you are focusing on?



The background of the slide is a high-angle aerial photograph of Earth. It shows a vast expanse of white, fluffy clouds covering most of the surface. In some areas, particularly towards the bottom, the dark blue and brownish-green colors of land and water are visible through the cloud cover. The sky at the very top is a clear, pale blue.

Design space

*what can vary in how you are thinking
about the problem?*

Framing the Right Design Problem!

No design problem exists fully formed.

A key job of a designer is to specify the exact problem that needs to be solved.



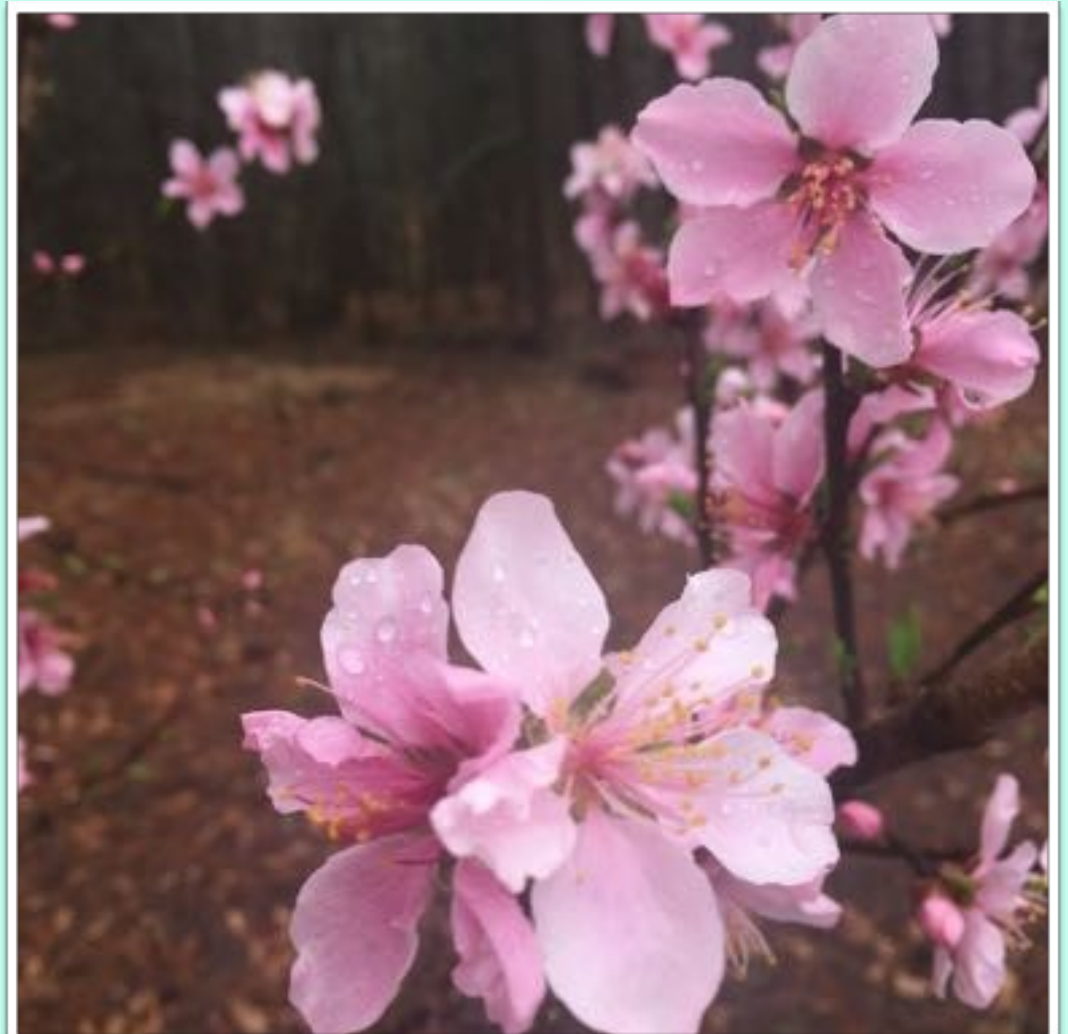
Framing is an Iterative Process

- *Project timeline changes*
- *Budget gets cut*
- *Designer's understanding evolves*



Needs

**You are NOT
your user!**



Personas



Personas are fictional characters that represent key types of technology's intended users that were discovered through formative work.

Key characteristics of personas

- **Based in Data**
- **Embody the various goals, traits, behaviors, and contexts that could influence if and how the technology is going to be used.**
- **Represent distinct classes of intended users**

Sample Persona

Mary



Demographics

Age: 49

Occupation: project manager

Location: Irvine, CA

Life stage: Divorced, one child

Motivators

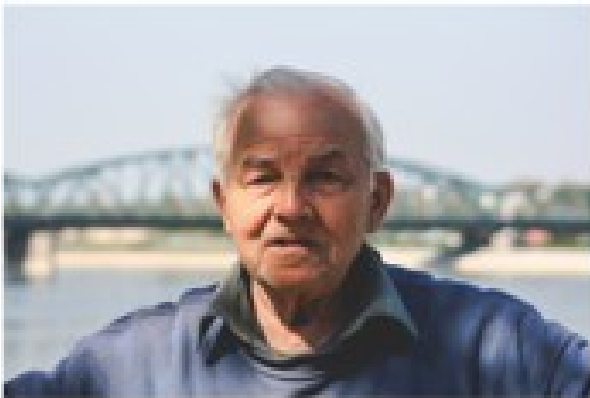
- Professional success
- Being a good mom to her 10-year-old daughter
- Novelty: Mary gets bored easily and has a high need for novel experiences
- Social contact: Mary is very social and enjoys working with people and spending social time with friends

Constraints

- Suburban living environment: Mary gets little activity in routine daily life
- Work days packed with meetings
- Free evening time limited: Mary often works in the evening, or wants to spend time with her daughter. Free on nights when daughter with her dad
- High technical literacy

Sample Persona

John



Demographics

Age: 68

Occupation: Mechanical engineer

Location: Ann Arbor, MI

Life stage: Retired, living with his wife

Motivators

- Staying healthy as long as possible
- Spending quality time with his grandchildren
- Being independent and not being burden on his children
- Continuing to learn

Constraints

- Weather: Often too cold or too hot to exercise outdoors
- Access to stationary bike at home
- Strong need for structure and routine
- Travels to spend time with grandchildren
- Moderate technical literacy



How to create Personas?

Elements of Personas

- **Demographics**

- *like age, technical literacy and*
- *if it's relevant, the kind of living environment that they're in*
- *Potential socioeconomic status.*
- *Issues that might affect how they would be using the technology*

- **goals and motivators.**

- *So these are issues related to how the technology would be used.*
- *The goals that these individuals will be pursuing using the technology.*
- *And motivators that could be leveraged to help them use the technology*
or
- *that might act as barriers to technology use.*

Elements of Personas

- **Behaviors**

- *such as routines and work practices are often included insofar as they're things to which technology can be integrated.*
- *Or again if they might act as barriers to technology use.*

- **Context,**

- *such as living and working environment and other aspects of physical and social context that might be relevant to determining technology use, so all of these can be integrated into personas.*

Benefits of Personas

- **Personalize and summarize a lot of data from the formative research**
- **Provide the mental shortcut for the design considerations**
 - *Would Mary do this?*
 - *How would John react to this particular feature?*
- **Allow discussion of and agreement on who the intended users are**
- **Easy to understand by non-designers**
- **Help check tendencies to design for oneself or the “elastic user”**

Some Final Considerations

- **Personas should feel real.**
 - *They really should accurately summarize available data.*
- **Personas should always try to focus on common rather than idiosyncratic characteristics.**
- **Content should be relevant**
 - *e.g. family situation might not be relevant for design of a workplace technology*

Design Scenarios



Scenarios



- **Scenarios are short stories about specific users who are using technology being designed to accomplish specific goals in a specific context.**

Scenarios bridge findings about user needs and ideas about how a technology may support those needs.



Sample Scenario

Ever since Mary returned to work after her heart attack last year, she has struggled to incorporate regular physical activity into her daily life. HealthyHeart, an app she recently discovered, is slowly changing that. As Mary is finishing her meeting, her activity tracker gently vibrates to indicate that HealthyHeart has sent her an activity suggestion. When Mary looks at her phone, she finds a message showing her a 1500-step walking route to her next meeting that goes by the beautiful fountain on her company's campus. She has just enough time to walk over, so she hits "thumbs-up" to indicate she liked the suggestion, and she follows the suggested route. That evening, after dinner, HealthyHeart alerts her that there is a salsa dance three days later at a restaurant 20 minutes away. "Oh, great!" Mary thinks. "This could be a fun way to get more intense activity to meet my weekly activity prescription." She taps "let's do it!" and salsa dancing, along with the directions, is added to her calendar.

What did we learn?

- **Mary's days are packed with meetings**
- **HealthyHeart is proactive**
- **Its suggestions are just-in-time and contextually tailored**
- **It helps Mary discover novel physical activities**
- **It's intended to induce feeling of pleasant surprise**

Why use scenarios?

- **Support reflection during the design process**
- **Open-ended and easily revised**
- **Anchor design discussion**

Some HealthyHeart Considerations

- **How many suggestions would Mary tolerate before getting annoyed?**
- **What would the system need to know to provide good suggestions?**
- **How should such a system respond if Mary doesn't like a suggestion?**
- **Where would ideas for novel activities come from?**

Elements of a scenario

- **Setting:**
 - *situation(state of the system, context of use) in which activity occurs*
- **Agent(s):**
 - *people performing the activity described in scenario*
- **Goals:**
 - *objectives agents are trying to accomplish*
- **Actions:**
 - *activities that agents do to accomplish their goals*
- **Events:**
 - *things that happen to users while doing the activities in the scenario*

Types of scenarios

- **Problem scenarios:**
 - *describe features of the current situation*
- **Goal or task-based scenarios:**
 - *what the user wants to do*
- **Activity scenarios:**
 - *transform current practice into new design features*

Types of scenarios

- **Full scale task scenarios:**
 - *steps to accomplish a task*
- **Information scenarios:**
 - *how users perceive, interpret, and make sense of information*
- **Interaction scenarios:**
 - *physical actions and system responses that respond to users' tasks*

Scenario Considerations

- **You don't need to be a great writer to write useful scenarios**
- **Scenarios should be quick. More rough scenarios are often more useful than fewer scenarios that are more refined**
- **Scenarios are process artifacts. Do only as much as is helpful**
- **Scenarios can help uncover unwanted technology effect**

Scenario summary

- **Can help test early design ideas**
- **Support communication**
- **Help designers reflect on intended and unintended consequences of their design decisions**

Storyboards



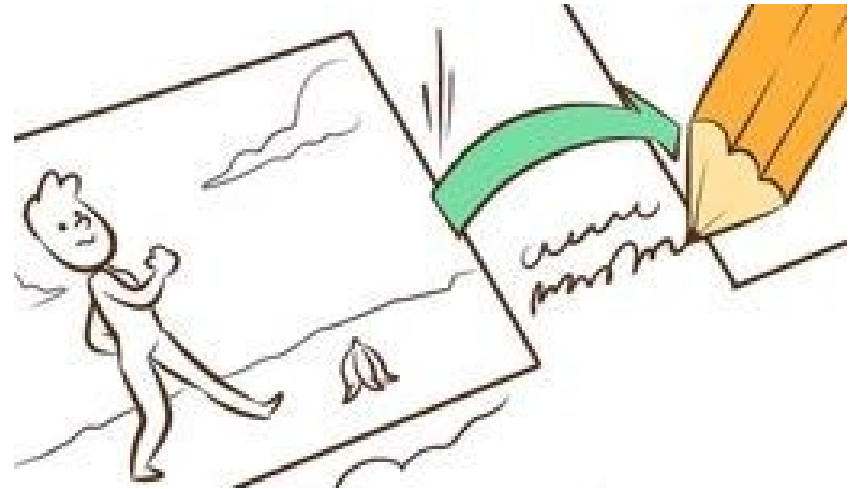
Storyboards

Visual presentation of stories about how individuals use the technology that's being developed to support their goals.

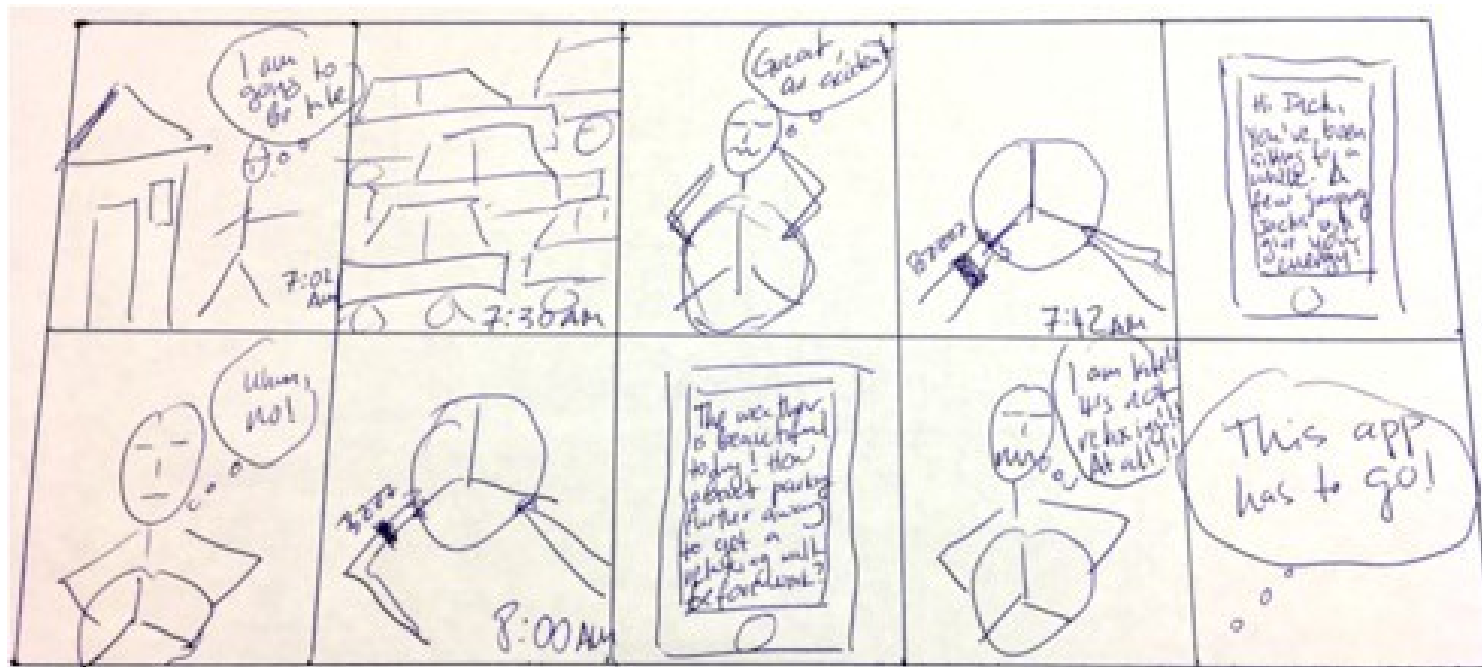


What is a Storyboard?

- A storyboard is a **series of sketches** that **visually conveys** how a **user engages in an activity** that involves the technology that is being developed.



This is a storyboard!



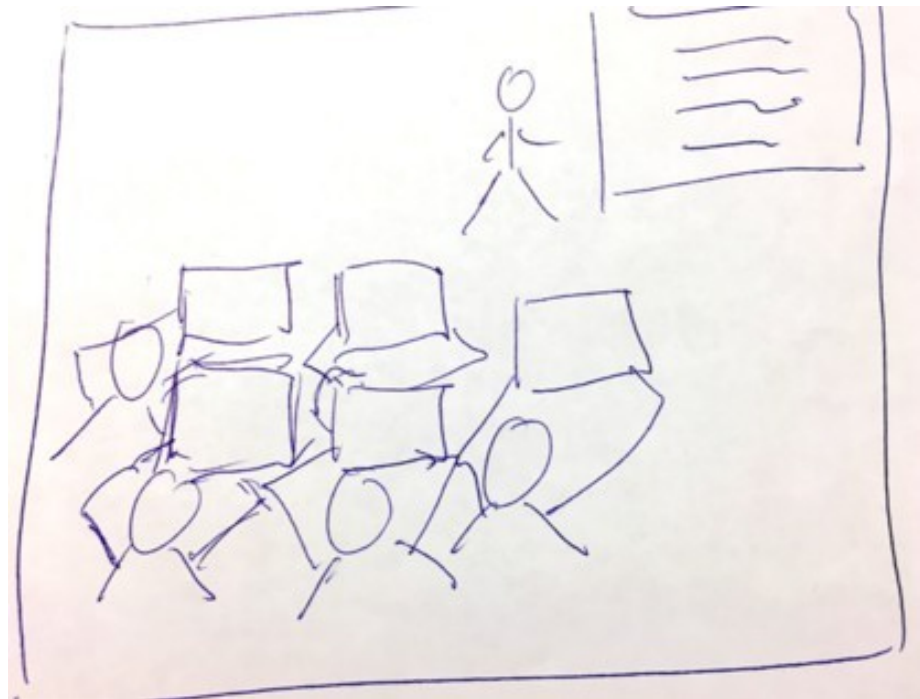
Lessons learnt?

- **If suggestions are not doable, it leads to user frustration**
- **If just-in-time suggestions are poorly timed, it can be really annoying**
- **Frustration accumulates. Enough of it, and the system might be abandoned**

Why storyboards? (If you have scenarios)

- **They help you think more deeply about...**
 - – *environments where the system is used*
 - – *physical constraints (size of system, space where it's used...)*
 - – *User's motivation and emotion*
 - – *relationships among multiple people*

Example!



Elements of a storyboard

- **Zoom:**
 - *how wide/narrow is the focus of the frame?*
- **Angle:**
 - *from whose point of view is the content in the frame? User's? Observer's?*
- **Detail:**
 - *what is the frame focused on?*
- **Emotion:**
 - *what emotion is the user expressing?*
- **Setting:**
 - *where does the action take place?*

Storyboard Considerations

- **How many panels are needed?**
- **What interactions will be represented?**
- **How will time be represented?**
- **How will the user (“character”) fit?**
- **What text will be used?**
- **Outline screens**
- **Add detail as needed**

Some Final thoughts on Storyboards

- **You don't need to be a great artist to draw useful storyboards**
- **Storyboarding quickly helps you create more storyboards**
- **Storyboards most helpful for conveying physical environment, emotion, and relationships**
- **Play to your strengths—often either a scenario or a storyboard will work**

Resources

- **Personas:**

- <https://unamo.com/blog/general/how-to-create-ux-personas>
- <https://uxmastery.com/create-ux-personas/>
- <https://www.uxpin.com/studio/blog/ux-designers-5-minute-guide-lean-personas/>
- <http://www.uxbooth.com/articles/creating-personas/>

- **Scenarios:**

- <https://www.usability.gov/how-to-and-tools/methods/scenarios.html>
- <https://www.interaction-design.org/literature/topics/user-scenarios>
- <https://www.webdesignerdepot.com/2015/04/how-to-perfect-your-ux-with-persona-scenarios/>

- **Storyboards:**

- *Storyboarding in UX Design*
 - <https://uxplanet.org/storyboarding-in-ux-design-b9d2e18e5fab>
 - <https://www.smashingmagazine.com/2017/10/storyboarding-ux-design/>
 - <https://www.spotless.co.uk/insights/three-tips-for-storyboards/>

Design Rationale - Making Design Decisions





Making Choices as a Designer!



Design Rationale

Design rationale is articulation and analysis of tradeoffs of different alternatives to guide design

Design Rationale

Goals: To understand the design options, their pros and cons, and to make principled decisions about which design options to pursue.



Questions, Options, Criteria (QOC Framework)

- A feature is represented by multiple possible **options** which answer a particular **question**
- **Criteria** help articulate their tradeoffs and guide the choice of which option(s) to go with

MacLean, A., Young, R.M., Bellotti, V. & Moran, T.P. (1991). Questions, options, and criteria: Elements of design space analysis. Human-Computer Interaction 6, 3-4: 201-250.



Diet Tracking Application

Questions

- **Provide structure to the design space**
- **Help uncover and define alternatives**

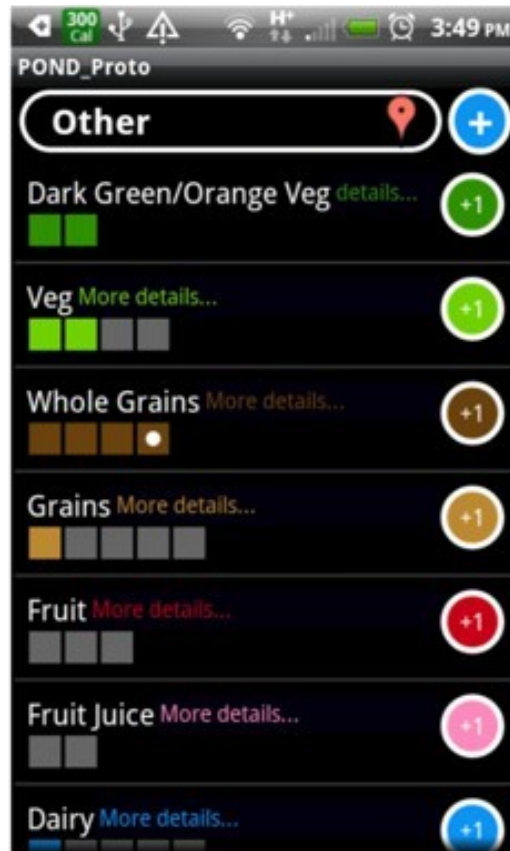
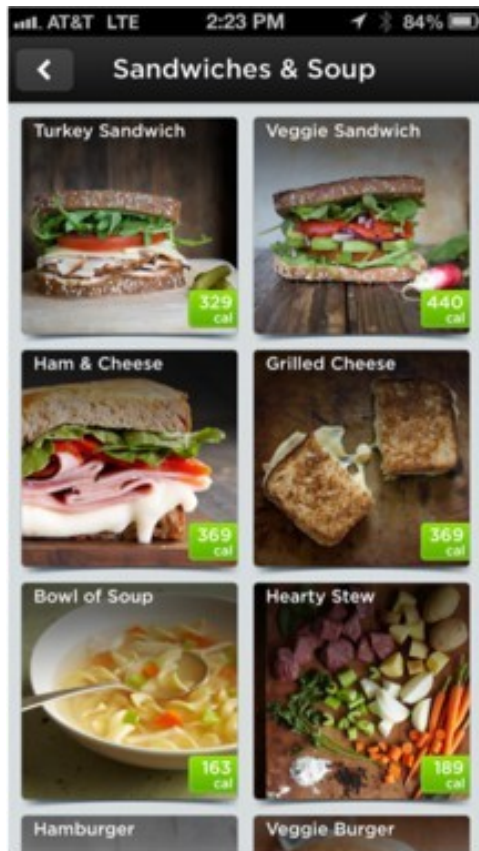
Example Questions

- **How should consumed food be entered into the food log?**
- **What kinds of food should be tracked?**
- **At what granularity should food be tracked?**
- **...**

Options

- **Different potential design answers to the same question**
- **How should consumed food be entered into the food log?**
 - – *By finding consumed food in a database*
 - – *By logging its nutritional content (grains, fruits...)*
 - – *By taking a picture of food*
 - – ...

Design Alternatives



Criteria

- **Required and desirable properties the design should satisfy**
- **Differ in importance and generality**
- **Help determine reasons for decisions**

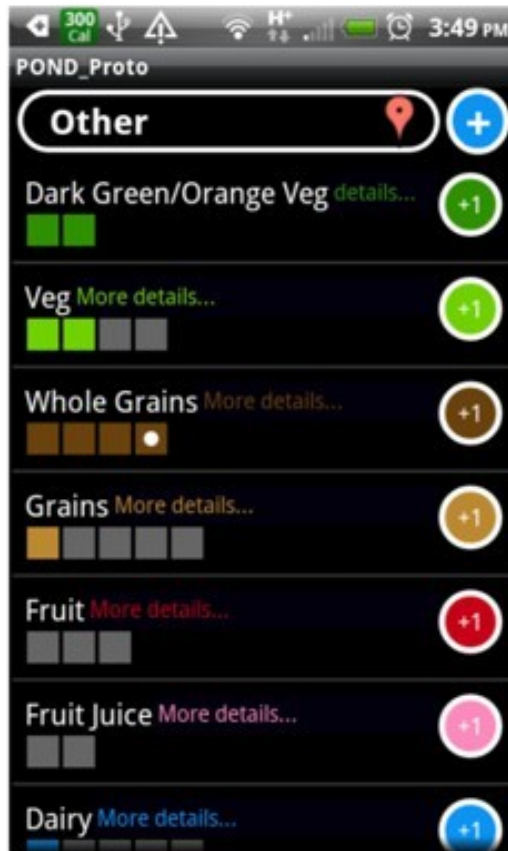
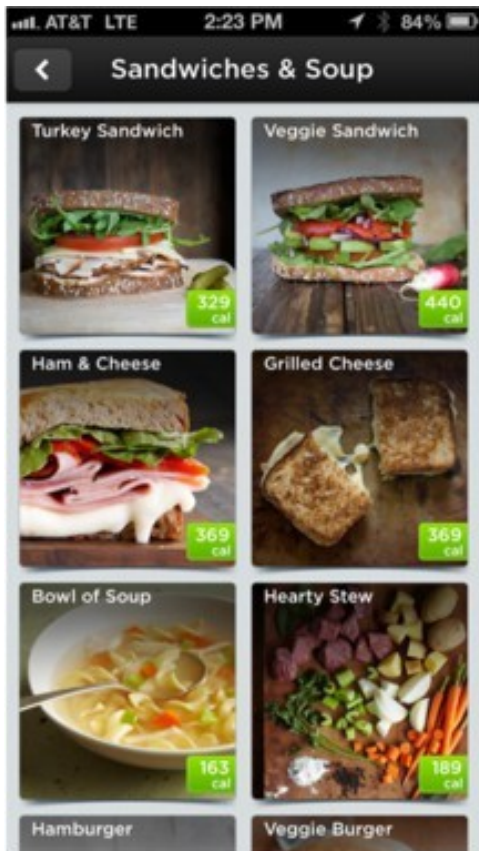
Sources of Criteria

- **Formative work with target users**
- **Usability broadly construed**
 - *e.g., integration with daily routines, privacy, social acceptability, user expectations, etc.*
- **Previous studies**
- **Behavioral theory**
- **Requirements from other parts of the design**

Food Tracking Criteria

- **Can be maintained long-term**
- **Doesn't require deep nutritional knowledge**
- **Fast to use**
- **Easy to use from the beginning**
- **Supports graphing and statistical analysis of data**

Evaluating Options using Criteria



Contenders : Narrowing Design Focus



Things to Consider

- **There might not be one clear answer**
- **It's important to know the most important criteria**
- **A design decision can affect many other aspects of the project**
- **Empirical data (from surveys, focusgroups, small studies) can inform design decisions**

Summary

- **Deciding which option to pursue is a key design activity**
- **Design decisions need to be intentional and reasoned**
- **QOC enables designers to make design decisions in a systematic way**